



C++ - Module 07

C++ templates

Summary:

This document contains the exercises of Module 07 from C++ modules.

Version: 10.1

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Chapter I

Introduction

C++ is a general-purpose programming language created by Bjarne Stroustrup as an extension of the C programming language, or "C with Classes" (source: [Wikipedia](#)).

The goal of these modules is to introduce you to **Object-Oriented Programming**. This will be the starting point of your C++ journey. Many languages are recommended to learn OOP. We decided to choose C++ since it's derived from your old friend C. Because this is a complex language, and in order to keep things simple, your code will comply with the C++98 standard.

We are aware modern C++ is way different in a lot of aspects. So if you want to become a proficient C++ developer, it's up to you to go further after the 42 Common Core!

Chapter II

General rules

Compiling

- Compile your code with `c++` and the flags `-Wall -Wextra -Werror`
- Your code should still compile if you add the flag `-std=c++98`

Formatting and naming conventions

- The exercise directories will be named this way: `ex00`, `ex01`, ... , `exn`
- Name your files, classes, functions, member functions and attributes as required in the guidelines.
- Write class names in **UpperCamelCase** format. Files containing class code will always be named according to the class name. For instance: `ClassName.hpp/ClassName.h`, `ClassName.cpp`, or `ClassName.hpp`. Then, if you have a header file containing the definition of a class "BrickWall" standing for a brick wall, its name will be `BrickWall.hpp`.
- Unless specified otherwise, every output message must end with a newline character and be displayed to the standard output.
- *Goodbye Norminette!* No coding style is enforced in the C++ modules. You can follow your favorite one. But keep in mind that code your peer evaluators can't understand is code they can't grade. Do your best to write clean and readable code.

Allowed/Forbidden

You are not coding in C anymore. Time to C++! Therefore:

- You are allowed to use almost everything from the standard library. Thus, instead of sticking to what you already know, it would be smart to use the C++-ish versions of the C functions you are used to as much as possible.
- However, you can't use any other external library. It means C++11 (and derived forms) and **Boost** libraries are forbidden. The following functions are forbidden too: `*printf()`, `*alloc()` and `free()`. If you use them, your grade will be 0 and that's it.

- Note that unless explicitly stated otherwise, the `using namespace <ns_name>` and `friend` keywords are forbidden. Otherwise, your grade will be -42.
- **You are allowed to use the STL only in Modules 08 and 09.** That means: no **Containers** (vector/list/map, and so forth) and no **Algorithms** (anything that requires including the `<algorithm>` header) until then. Otherwise, your grade will be -42.

A few design requirements

- Memory leakage occurs in C++ too. When you allocate memory (by using the `new` keyword), you must avoid **memory leaks**.
- From Module 02 to Module 09, your classes must be designed in the **Orthodox Canonical Form, except when explicitly stated otherwise**.
- Any function implementation put in a header file (except for function templates) means 0 to the exercise.
- You should be able to use each of your headers independently from others. Thus, they must include all the dependencies they need. However, you must avoid the problem of double inclusion by adding **include guards**. Otherwise, your grade will be 0.

Read me

- You can add some additional files if you need to (i.e., to split your code). As these assignments are not verified by a program, feel free to do so as long as you turn in the mandatory files.
- Sometimes, the guidelines of an exercise look short but the examples can show requirements that are not explicitly written in the instructions.
- Read each module completely before starting! Really, do it.
- By Odin, by Thor! Use your brain!!!



Regarding the Makefile for C++ projects, the same rules as in C apply (see the Norm chapter about the Makefile).



You will have to implement a lot of classes. This can seem tedious, unless you're able to script your favorite text editor.



You are given a certain amount of freedom to complete the exercises. However, follow the mandatory rules and don't be lazy. You would miss a lot of useful information! Do not hesitate to read about theoretical concepts.

Chapter IV

Exercise 00: Start with a few functions

	Exercise: 00
Start with a few functions	
Directory: <i>ex00/</i>	
Files to Submit: <code>Makefile</code> , <code>main.cpp</code> , <code>whatever.{h, hpp}</code>	
Forbidden: <code>None</code>	

Implement the following function templates:

- **swap**: Swaps the values of two given parameters. Does not return anything.
- **min**: Compares the two values passed as parameters and returns the smallest one. If they are equal, it returns the second one.
- **max**: Compares the two values passed as parameters and returns the greatest one. If they are equal, it returns the second one.

These functions can be called with any type of argument. The only requirement is that the two arguments must have the same type and must support all the comparison operators.



Templates must be defined in the header files.

Running the following code:

```
int    main( void ) {

    int a = 2;
    int b = 3;

    ::swap( a, b );
    std::cout << "a = " << a << ", b = " << b << std::endl;
    std::cout << "min( a, b ) = " << ::min( a, b ) << std::endl;
    std::cout << "max( a, b ) = " << ::max( a, b ) << std::endl;

    std::string c = "chaine1";
    std::string d = "chaine2";

    ::swap( c, d );
    std::cout << "c = " << c << ", d = " << d << std::endl;
    std::cout << "min( c, d ) = " << ::min( c, d ) << std::endl;
    std::cout << "max( c, d ) = " << ::max( c, d ) << std::endl;

    return 0;
}
```

Should output:

```
a = 3, b = 2
min(a, b) = 2
max(a, b) = 3
c = chaine2, d = chaine1
min(c, d) = chaine1
max(c, d) = chaine2
```

Chapter V

Exercise 01: Iter

	Exercise: 01
Iter	
Directory: <i>ex01/</i>	
Files to Submit: <code>Makefile</code> , <code>main.cpp</code> , <code>iter.{h, hpp}</code>	
Forbidden: None	

Implement a function template `iter` that takes 3 parameters and returns nothing.

- The first parameter is the address of an array.
- The second one is the length of the array, passed as a `const` value.
- The third one is a function that will be called on every element of the array.

Submit a `main.cpp` file that contains your tests. Provide enough code to generate a test executable.

Your `iter` function template must work with any type of array. The third parameter can be an instantiated function template.

The function passed as the third parameter may take its argument by `const` reference or non-`const` reference, depending on the context.



Think carefully about how to support both `const` and non-`const` elements in your `iter` function.

Chapter VI

Exercise 02: Array

	Exercise: 02
Array	
Directory: <i>ex02/</i>	
Files to Submit: <code>Makefile</code> , <code>main.cpp</code> , <code>Array.{h, hpp}</code> and optional file: <code>Array.hpp</code>	
Forbidden: None	

Develop a class template **Array** that contains elements of type **T** and that implements the following behavior and functions:

- Construction with no parameter: Creates an empty array.
- Construction with an unsigned int **n** as a parameter: Creates an array of **n** elements initialized by default.
Tip: Try to compile `int * a = new int();` then display `*a`.
- Construction by copy and assignment operator. In both cases, modifying either the original array or its copy after copying musn't affect the other array.
- You **MUST** use the operator `new[]` to allocate memory. Preventive allocation (allocating memory in advance) is forbidden. Your program must never access non-allocated memory.
- Elements can be accessed through the subscript operator: `[]`.
- When accessing an element with the `[]` operator, if its index is out of bounds, an `std::exception` is thrown.
- A member function `size()` that returns the number of elements in the array. This member function takes no parameters and must not modify the current instance.

As usual, ensure everything works as expected and turn in a `main.cpp` file that contains your tests.

Chapter VII

Submission and peer-evaluation

Turn in your assignment in your `Git` repository as usual. Only the work inside your repository will be evaluated during the defense. Don't hesitate to double check the names of your folders and files to ensure they are correct.

During the evaluation, a brief **modification of the project** may occasionally be requested. This could involve a minor behavior change, a few lines of code to write or rewrite, or an easy-to-add feature.

While this step may **not be applicable to every project**, you must be prepared for it if it is mentioned in the evaluation guidelines.

This step is meant to verify your actual understanding of a specific part of the project. The modification can be performed in any development environment you choose (e.g., your usual setup), and it should be feasible within a few minutes — unless a specific timeframe is defined as part of the evaluation.

You can, for example, be asked to make a small update to a function or script, modify a display, or adjust a data structure to store new information, etc.

The details (scope, target, etc.) will be specified in the **evaluation guidelines** and may vary from one evaluation to another for the same project.